

drives. But that doesn't mean that there aren't variations in the individual performances of SSDs, even if they have the same SATA. The performance is affected by the SSD controller or the chip, which is responsible for managing the data movement in the memory. Some controllers prove faster than others while processing certain kinds of data. Random access rates also play an important role here when the user is running programs. These rates are measured in IOPS units. 40,000 IOPS or above indicates an efficient drive. Please note that you cannot access these rates for all applications; however, knowing what is possible with your SSD can help you make a better choice.

Form factor

Generally, manufacturers of SSDs follow a standard form factor of 6.35cm (2.5 inches). However, their thickness may vary between 7mm and 9.5mm. Some SSDs are as thick as 12mm, but they do not fit the normal tight enclosures. So before you make a choice, do take a look at the form factor, and ensure that it matches your server's enclosures.

TRIM support

Most of us know that even after deleting a file, it does not get permanently deleted. It still resides on the drive, though users

are unable to locate it. A pile up of such files can hamper the performance of an SSD. So this is when TRIM support becomes important. With it, users get to decide on what files they can do away with permanently.

Warranty

SSDs do undergo wear and tear over a longer span of time. If you are aware of the kind of usage your server will be subjected to and the fact that it will undergo constant wear and tear, you can choose an SSD that comes with better warranty terms and a longer lifespan. Most SSDs today come with a three or four-year warranty, ensuring some peace of mind for an IT head.

Which flash memory technology to choose: MLC or SLC?

MLC and SLC are the two flash memory technologies that one can choose from while buying an SSD. While MLC stands for multi-level cell and stores multiple bits per cell, SLC stands for single level cell and stores only one bit per cell. SSDs with MLC technology are comparatively cheaper than those with SLC technology, but the latter are faster, longer lasting and more reliable.

Popular SSDs available in the market

Samsung SSD 840 Pro Series



Price: ₹ 26,875

- Brand: Samsung
- Model: MZ-7PD512BW
- Hard drive capacity: 512GB
- Warranty: Five years from Samsung

Kingston HyperX 3K SSD



Price: ₹ 26,790

- Brand: Kingston
- Hard drive capacity: 480GB
- High-speed SATA Rev 3.0 (6Gbps) transfer speeds
- Warranty: Three years' manufacturer's warranty